Oracle EPM Predictive Cash Forecasting

Driver-Based Forecasting - Advanced Lab Guide

Product Design Documentation • Implementation Guide • Training Material • Technical Reference • Troubleshooting Handbook

Advanced-stage focus

Design, configure and operate driver-based cash forecasting using Oracle PCF's eleven driver methods, blending operational assumptions with revenue, expense, payroll, project, capital and tax data to create explainable rolling cash forecasts.

Lab Property	Design
Primary horizon	13-week periodic rolling forecast with daily/periodic concepts
Primary methods	11 Oracle driver-based methods plus blended forecast-method design
Core inputs	Revenue, expense, projects, payroll, capex, taxes, DSO, DPO and pay terms
Core outputs	Operating, investing and financing cash impacts; closing cash; headroom
Audience	Advanced Oracle EPM consultants, treasury analysts, solution architects

BISP Trainings

1. Purpose and Advanced Learning Outcomes

Why driver-based forecasting matters

Driver methods translate operational business plans into cash timing using explicit assumptions. Unlike a purely historical trend, the forecast can explain why cash occurs, when it occurs, and which business assumption drives it.

- Configure assumptions by entity, line item and enabled custom dimensions.
- Load or input driver data and execute Daily Process Forecast / Periodic Process Forecast rules.
- Apply Revenue, Project, DSO, Expense, Fixed Asset, Recurring, Salary, Project Payment, Direct Tax, Indirect Tax and DPO methods.
- Blend methods across line items, entities and forecast period ranges.
- Validate pay-term splits, dates, DSO/DPO assumptions and driver coverage.
- Measure forecast accuracy, bias and scenario sensitivity.
- Build a production runbook for automation, exceptions, approval and troubleshooting.

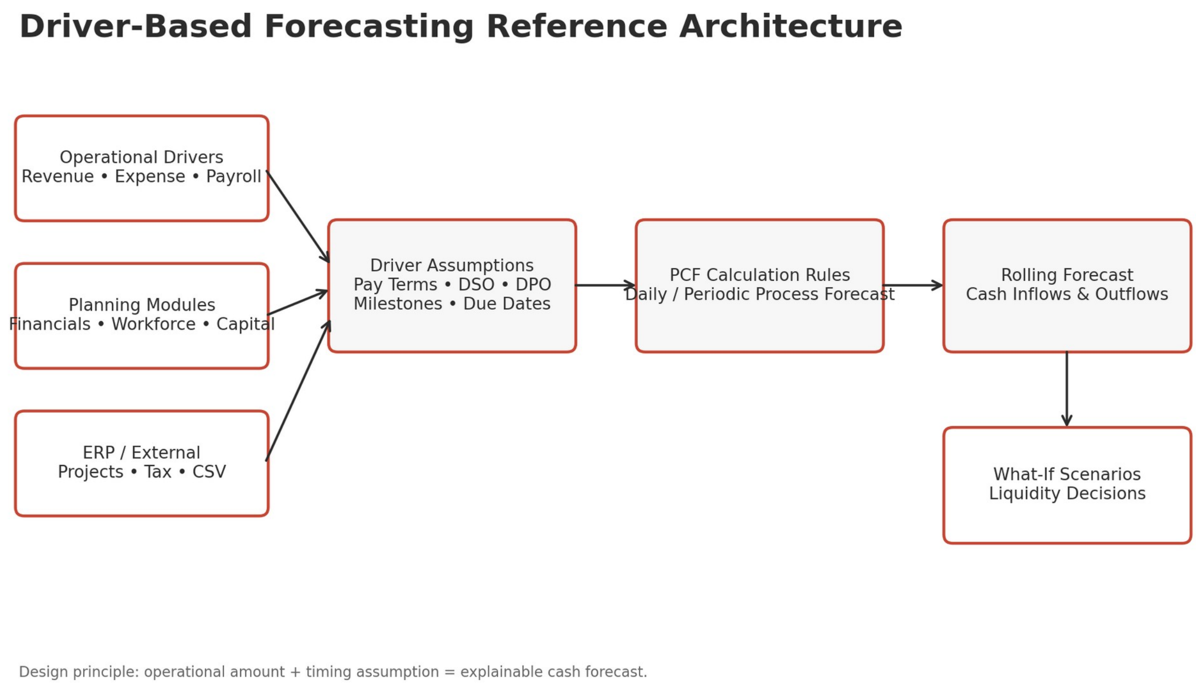
2. Current Oracle Forecast-Method Framework

Forecast method	Best use	Design note
Driver Based	Operational amounts plus explicit timing assumptions	Oracle provides 11 driver-based methods.
Smart Drivers	Near-term Fusion ERP transactional data	Useful especially for early daily/periodic periods.
Trend Based	Stable historical cash patterns	Periodic forecasts only.
Prediction Based	Historical time-series based cash prediction	Predictive Planning / AutoPredict use cases.
Manual Input	One-off or hard-to-model items	Use with governance and commentary.

Blended design

Oracle PCF supports different forecast methods by line item, period range, entity and version. A mature implementation normally blends methods rather than forcing one method across the entire forecast.

3. Driver-Based Forecasting Architecture



Process Contract

Layer	Responsibility
Driver source	Provide forecast revenue, expense, salary, project, capital or tax values.
Assumption layer	Define pay terms, DSO/DPO, milestones, due dates, recurring schedules.
Calculation layer	Calculate and post cash into the appropriate daily or periodic bucket.
Rolling forecast	Consolidate line-item results into cash inflows/outflows and balances.
Scenario layer	Change assumptions and compare liquidity outcomes.

4. The Eleven Driver-Based Methods

#	Method	Cash direction	Typical driver
1	Revenue Receipts	Inflow	Revenue + pay terms
2	Project Receipts	Inflow	Project revenue + milestones + pay terms
3	DSO Receipts	Inflow	Outstanding receivables/revenue + average DSO
4	Expense Payments	Outflow	Expense + pay terms
5	Fixed Asset Payments	Outflow / Investing	Capex spend + pay terms
6	Recurring Payments	Outflow	Recurring amount + frequency
7	Salary Payments	Outflow	Salary expense + salary incidence / pay terms
8	Project Payments	Outflow	Project expense + milestones + pay terms
9	Direct Tax Payments	Outflow	Tax liability + installment % + due dates
10	Indirect Tax Payments	Outflow	Tax liability + basis + due dates / pay terms
11	DPO Payments	Outflow	Outstanding expense + average DPO

5. Revenue Receipts - Revenue + Pay Terms

Use case

Use when product or service revenue follows a reasonably stable payment pattern, for example 70% collected three days after sale and 30% five days after sale.

Input	Example
Revenue driver	$10.0M
Pay Term 1	70% due in 3 days
Pay Term 2	30% due in 5 days
Forecast result	$7.0M on Day +3; $3.0M on Day +5

Step-by-step Calculation

1. Load or input Revenue = $10.0M for the source period.
2. Validate pay-term percentages total 100%.
3. Calculate $10.0M × 70% = $7.0M and post to Day +3.
4. Calculate $10.0M × 30% = $3.0M and post to Day +5.
5. If a due period falls outside the active forecast range, the amount is not posted in that range.

6. Project Receipts - Milestones + Pay Terms

Input	Example
Project contract value	$20.0M
Milestone 1	40% at W2
Milestone 2	60% at W8
Pay terms	50% on milestone date, 50% 14 days later

1. Milestone 1 amount = $20.0M × 40% = $8.0M.
2. Cash from Milestone 1 = $4.0M at W2 + $4.0M two weeks later.
3. Milestone 2 amount = $20.0M × 60% = $12.0M.
4. Apply the same pay-term pattern to the W8 milestone, subject to forecast-horizon boundaries.

Best fit

Engineering, construction, real estate, consulting and contract-based businesses where cash follows milestones rather than straight-line revenue.

7. DSO Receipts - Outstanding Revenue + Average DSO

Input	Example
Outstanding Revenue	$12.0M
Average DSO	42 days
Adjusted DSO in Stress	50 days
Driver level	Entity or Party

Conceptually, the driver shifts the expected cash receipt based on the average number of days it takes to collect revenue. This is useful for future periods where detailed customer invoices do not yet exist.

Scenario insight

Increasing DSO from 42 to 50 days does not change revenue, but it pushes cash later, directly reducing near-term liquidity.

8. Expense Payments - Expense + Pay Terms

Expense category	Forecast expense	Pay terms	Cash timing
Utilities	$0.60M	100% in 15 days	$0.60M shifted 15 days
Travel	$0.40M	50% in 7 days / 50% in 30 days	$0.20M + $0.20M
Professional Fees	$1.00M	100% in 45 days	$1.00M shifted 45 days

Design rule

Keep the expense plan separate from the cash timing assumption. Changing pay terms should move cash without changing the underlying operating expense forecast.

9. Fixed Asset Payments - Capex + Pay Terms

Input	Example
Fixed Asset Spend	$6.0M
Asset Class	Manufacturing Equipment
Pay Term 1	20% advance
Pay Term 2	60% on delivery in W6
Pay Term 3	20% 30 days after delivery

1. Advance payment = $6.0M × 20% = $1.2M.
2. Delivery payment = $6.0M × 60% = $3.6M in W6.
3. Final payment = $6.0M × 20% = $1.2M 30 days later.
4. Classify the cash flow under Investing Activities rather than Operating Activities.

10. Recurring Payments

Driver	Example
Recurring amount	$0.25M
Pay Basis	Monthly
Start period	Sep-2026
Recurring frequency	Every month
Occurrences	6

Typical line items

Lease payments, rent, insurance and other contracted recurring expenses. The recurring schedule determines when the driver amount posts to forecast periods.

11. Salary Payments

Driver	Example
Annual salary and benefits	$24.0M
Salary basis	Annual
Incidence	Semi-monthly
Annual variable compensation	$3.0M
Variable pay due date	15-Mar-2027

For regular salary, the model converts the annual amount to the configured salary incidence. Separate annual or variable payments can be posted to their defined due date. In a weekly model, Oracle posts salary and earnings to the last day of the corresponding week under the documented logic.

12. Project Payments

Input	Example
Project expense	$10.0M
Milestone 1	30% at W3
Milestone 2	70% at W9
Pay terms	25% at milestone, 75% 30 days later

1. W3 milestone amount = $10.0M × 30% = $3.0M.
2. Immediate cash = $3.0M × 25% = $0.75M.
3. Deferred cash = $3.0M × 75% = $2.25M 30 days later.
4. Repeat the logic for W9 milestone, subject to the active forecast horizon.

13. Direct Tax Payments

Input	Example
Annual tax liability	$8.0M
Installment 1	25% due 15-Jun
Installment 2	25% due 15-Sep
Installment 3	25% due 15-Dec
Installment 4	25% due 15-Mar

Advanced behavior

If annual tax liability changes during the year, later installments can absorb the incremental or reduced liability after considering prior tax paid. This is more sophisticated than simply repeating the original installment amount.

14. Indirect Tax Payments

Input	Example
GST / VAT liability	$1.2M monthly
Tax Basis	Monthly
Payment incidence	Next period
Pay term	100% 20 days after period end

Indirect tax methods can combine tax basis, payment incidence and pay terms. The design should reflect statutory due dates and whether the payment occurs in the same or next period.

15. DPO Payments - Outstanding Expense + Average DPO

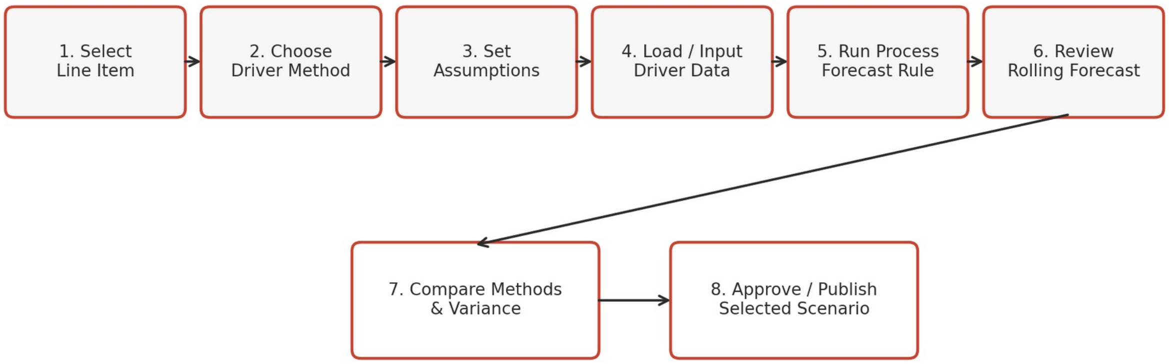
Input	Base	Stress
Outstanding Expense	$9.0M	$9.0M
Average DPO	38 days	45 days
Business interpretation	Normal payment behavior	Supplier payments deliberately extended

Best fit

Use DPO where pay terms are dynamic, where Smart Driver detail is unavailable, or for future periods beyond currently booked supplier invoices.

16. Operating Process

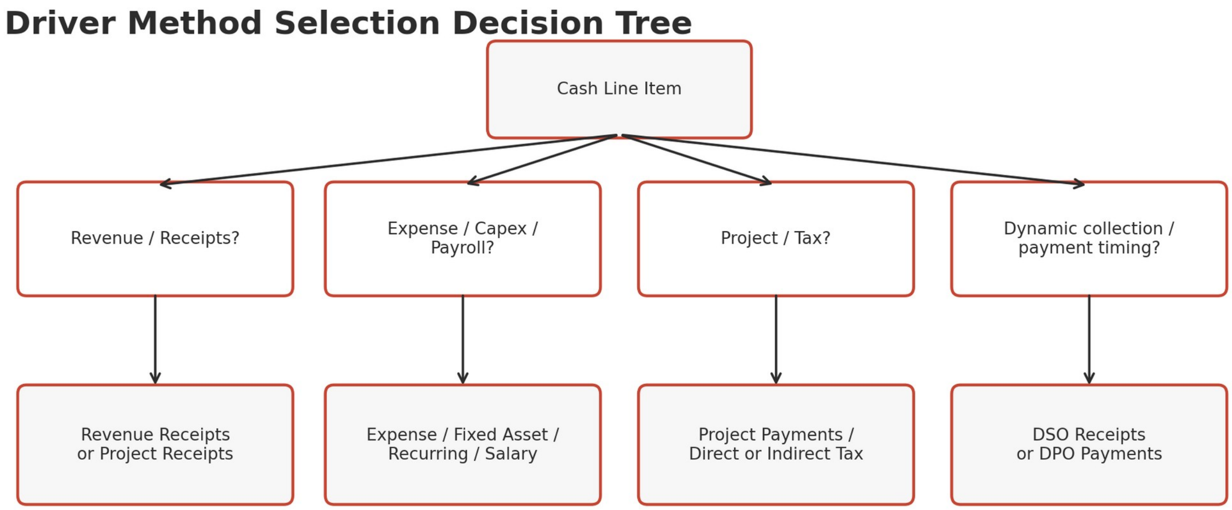
Driver-Based Forecasting Operating Process



Cash Manager Responsibilities

1. Set up assumptions such as pay terms, due dates, DSO/DPO and milestones.
2. Load or input the driver data used to calculate cash.
3. Run Daily Process Forecast or Periodic Process Forecast.
4. Review calculated inflows/outflows and confirm they post to the expected periods.
5. Compare scenario and method results before approving the rolling forecast.

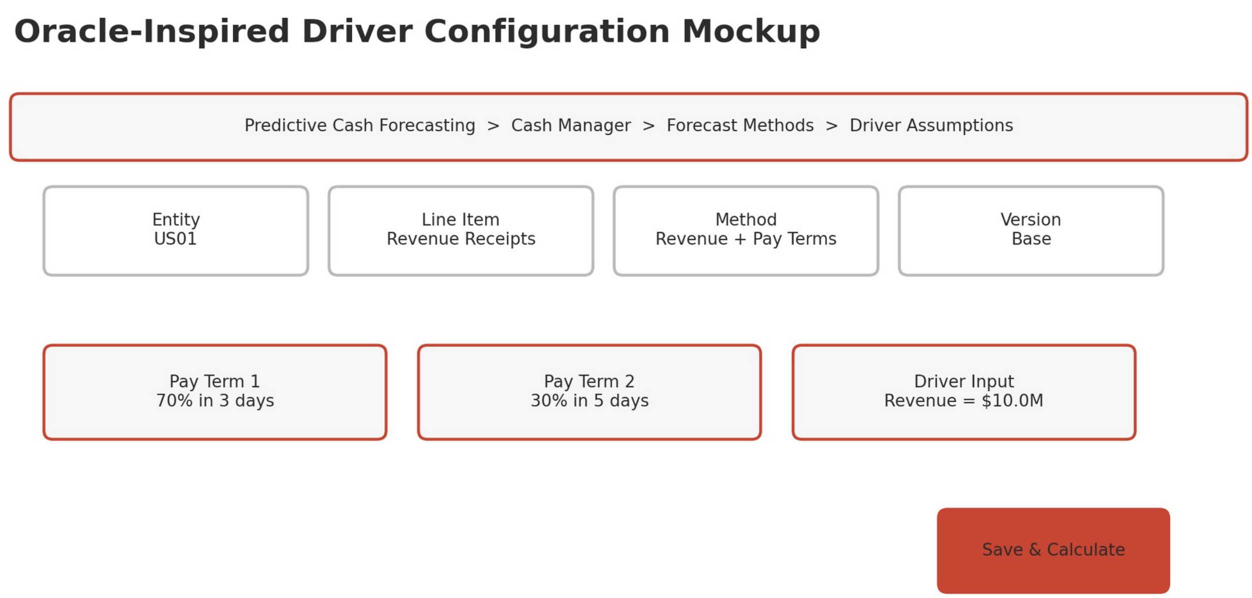
17. Method Selection Decision Tree



Method governance

Method selection should be explicit by line item. Document the reason, source availability, expected accuracy and fallback method for each material cash-flow line.

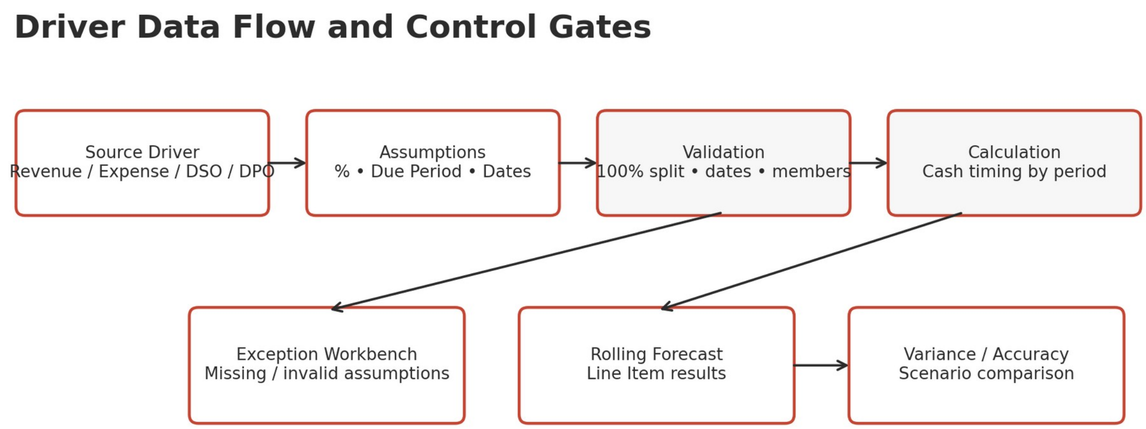
18. Oracle-Inspired Configuration Mockup



Configuration Checklist

- Enable Driver Based Forecasting during application configuration.
- Confirm sample line items and forms created by enabled methods.
- Configure assumptions by entity, line item and custom dimensions.
- Validate 100% pay-term allocations where the business rule requires full allocation.
- Test daily and periodic calculation behavior for horizon edges.

19. Data Flow and Validation Controls



Validation	Control
Pay-term percentage	Must reconcile to intended allocation, commonly 100%.
Due date / due period	Valid and inside forecast range when cash is expected to post.
DSO / DPO	Available at correct entity/party/period grain.
Driver amount	Reconciles to source plan or operational system.
Scenario / version	Results written to intended forecast version.
Mapping	No invalid entity, project, party, asset class or line item members.

20. Advanced 13-Week Integrated Lab

Driver	Assumption	13-week cash effect ($M)
Revenue Receipts	$18M revenue; 60% in 7d / 40% in 21d	18.00 inflow
DSO Future Revenue	$6M outstanding; DSO 42d	6.00 inflow shifted later
Expense Payments	$8M expenses; 50% 15d / 50% 30d	8.00 outflow
Salary Payments	$1.2M per week	15.60 outflow
Fixed Asset Payments	$5M capex; 20/60/20 schedule	5.00 investing outflow
Direct Tax	$2.4M installment in W8	2.40 outflow

Lab objective

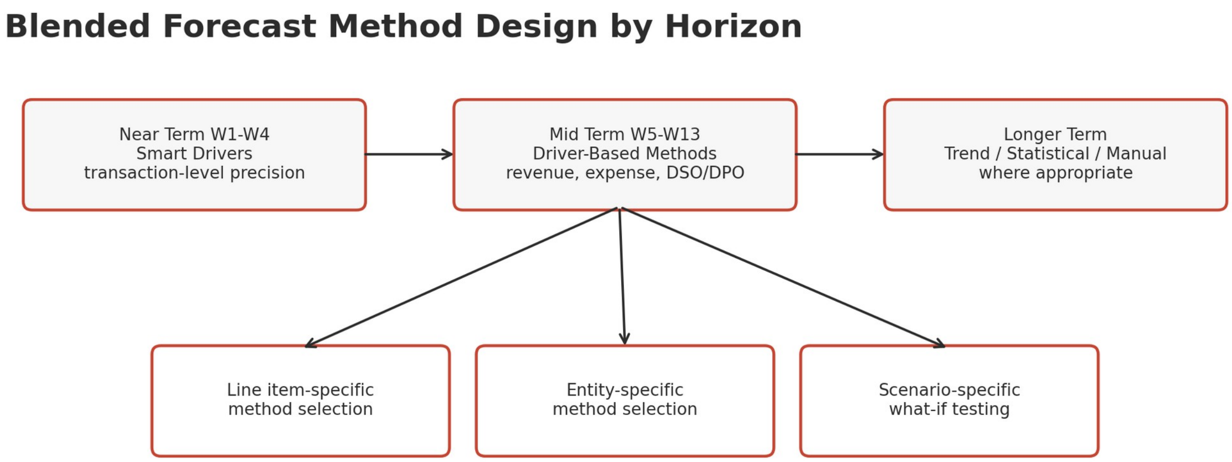
Build the Base forecast, then change DSO, DPO/pay terms and capex timing in Stress and Management Action scenarios to measure liquidity sensitivity.

21. Lab - Base vs Stress Scenario

Assumption	Base	Stress	Liquidity effect
Revenue pay pattern	60% 7d / 40% 21d	40% 7d / 60% 30d	Receipts shift later
Average DSO	42 days	50 days	Future receipts shift later
Expense pay terms	50% 15d / 50% 30d	75% 15d / 25% 30d	More near-term outflow
Capex delivery	W7	W5	Earlier investing outflow
Salary	No change	No change	Stable recurring burden

Metric	Base ($M)	Stress ($M)
Minimum Closing Cash	6.4	2.8
Minimum Buffer	3.0	3.0
Worst Headroom	3.4	-0.2
Funding Action	None	Revolver / timing actions

22. Blending Forecast Methods by Horizon



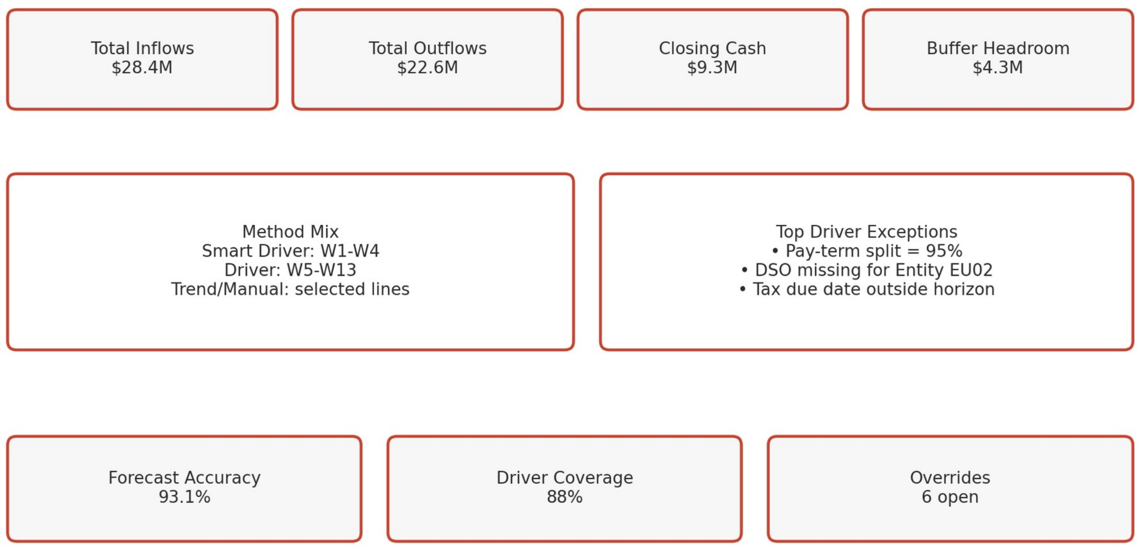
A robust design often uses Smart Drivers for the early transactional horizon, driver methods for the mid-term operational horizon, and trend/statistical/manual methods selectively for longer horizons or exceptional line items.

Example

Receivables may use Smart Drivers in W1-W4, Revenue Receipts or DSO in W5-W13, and statistical or trend methods beyond that - with method choices varying by entity and version.

23. Dashboard and Forecast Accuracy

Driver-Based Forecasting Dashboard Mockup



KPI	Interpretation
Driver Coverage	Percent of forecast explained by governed driver methods.
Forecast Accuracy	How closely forecasted cash matches actual cash.
Bias	Persistent over- or under-forecast direction.
Method FVA	Whether a driver method improves results versus a simpler baseline.
Buffer Headroom	Closing Cash less minimum cash buffer.
Override Count	Number of manual changes requiring governance review.

24. Error Logs and Troubleshooting

Driver Forecast Log / Error Examples

ERROR DRV-201 Pay-term percentages total 95%. Expected 100%. Forecast not calculated.

ERROR DRV-317 Average DSO missing for Entity EU02 / Party Retail.

WARN   DRV-122 Fixed Asset due period falls outside active forecast horizon; amount not posted.

INFO   DRV-001 Periodic Process Forecast completed: 146 driver intersections calculated.

Symptom	Likely cause	Resolution
Cash not posted	Due date falls outside forecast horizon	Validate horizon and assumptions.
Forecast too low/high	Pay-term split or driver amount incorrect	Reconcile source driver and percentages.
DSO/DPO result missing	Driver assumption missing at required grain	Load or enter entity/party period assumptions.
Salary timing wrong	Salary basis/incidence misconfigured	Correct basis and due-date/incidence settings.
Capex in wrong section	Incorrect line-item classification	Map Fixed Asset Payments to Investing Activities.
Scenario results overwritten	Wrong version/context used	Review POV and calculation rule target.

25. Security, Workflow and Governance

Role	Responsibilities
Administrator	Enable methods, dimensions, forms, rules and integrations.
Cash Manager	Maintain driver assumptions and evaluate forecast methods.
Controller	Review driver data, reconcile source values and approve forecast.
Treasurer	Use scenario outcomes for funding and liquidity actions.
Auditor	Review assumption changes, overrides, approvals and run evidence.

Audit evidence

Capture who changed the driver, old value, new value, entity, line item, period, scenario/version, business reason, approval status and calculation run ID.

26. Production Runbook and Implementation Checklist

☐ Enable only the driver methods required by the business design.

☐ Define the source and owner for every driver input and assumption.

☐ Load and reconcile revenue, expense, salary, project, capex and tax drivers.

☐ Configure DSO/DPO at the right entity/party granularity.

☐ Validate pay-term allocations, due periods, dates and forecast-horizon boundaries.

☐ Run Daily/Periodic Process Forecast and reconcile calculated cash flows.

☐ Build Base, Stress and Management Action versions.

☐ Compare forecast methods using accuracy, bias and FVA.

☐ Configure workflow for assumption changes and manual overrides.

☐ Test invalid assumptions, missing drivers, out-of-range due dates and restart scenarios.

☐ Complete performance testing and production scheduling.

☐ Publish support runbook and escalation matrix.

27. Advanced Student Challenge

Challenge scenario

A global manufacturer wants W1-W4 forecast precision using ERP transactions, W5-W13 forecast from operational plans, and a 6-month periodic extension. AR has dynamic customer behavior, AP has predictable DPO by supplier type, capex is milestone-based, payroll is stable, and annual tax is paid in four installments.

- Choose the forecast method for each line item and horizon.
- Define the driver inputs and assumptions required for every selected driver method.
- Design Base, Stress and Management Action scenarios.
- Explain how forecast accuracy and FVA would be measured by method.
- Define exception thresholds and governance for manual overrides.

28. Technical Reference and Completion Criteria

Oracle driver method	Key design inputs
Revenue Receipts	Revenue + pay-term percentages + due periods
Project Receipts	Project revenue + milestones + pay terms
DSO Receipts	Outstanding Revenue + Average DSO
Expense Payments	Expense + pay terms
Fixed Asset Payments	Fixed Asset Spend + pay terms
Recurring Payments	Amount + basis + start + frequency + occurrences
Salary Payments	Salary expense + basis + incidence + due date/pay terms
Project Payments	Project expense + milestones + pay terms
Direct Tax Payments	Tax liability + installment % + due dates
Indirect Tax Payments	Tax liability + basis + incidence + pay terms
DPO Payments	Outstanding Expense + Average DPO

Completion criteria

The student can explain, configure, calculate, reconcile and troubleshoot each driver method; blend methods by line item/entity/horizon; quantify liquidity outcomes; and operate the solution with production-grade controls.

Reference grounding: Oracle Predictive Cash Forecasting documentation current in 2026. Validate release-specific behavior in the target EPM environment before implementation.